Graphical Method for Finding Relationships Between Data

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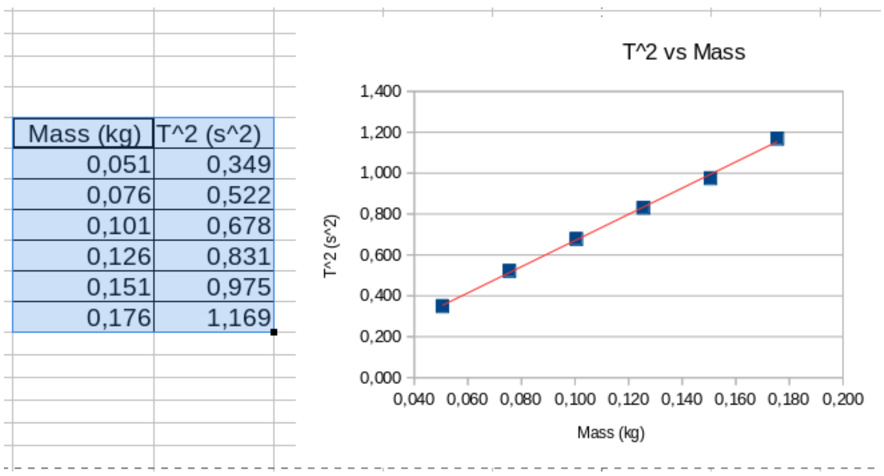
1 Graphs

The relationship between data recorded in columns in a spreadsheet is not immediately obvious. The method widely used to determine such relationships is the graphical method, which gives a pictorial view of the data. This is illustrated in the Figure 1 below.



Mass (kg)	T^2 (s^2)
0,051	0,349
0,076	0,522
0,101	0,678
0,126	0,831
0,151	0,975
0,176	1,169

Figure 1: A spreadsheet in which mass and time squared (T^2) data are recorded in two columns is shown on the left and graph of that data appears on the right.

By looking at the data in the spreadsheet it is not easy describe the relationship between the dependent variable (T^2 in Figure 1) and the independent variable ($Mass$). However, graphing the data reveals that T^2 increases as $Mass$ increases. They have a linear relationship.

In most laboratory experiments, the experimenter (you in this case) tends to vary one physical quantity at a time and measures another quantity which the you suspects is related to the first quantity.

The quantity that you can change is known as the independent variable. The quantity that changes in response to varing the independent variable is known as the dependent variable. The relationship between the two variables is best determined from a graph if the independent variable is plotted on the x-axis and the dependent variable is plotted on the y-axis.

2 Analysis and Interpretation of Graphs

It is important for you to be able to interpret relationship between variables on a graph and express it as a mathematical equation.
If points lie on a straight line, there is a linear relationship between variables. The mathematical equation of a straight line is

$$y = mx + c. \tag{1}$$

Important information can also be obtained from the slope and intercept, which are easily determined from a line of best fit through linearised data as follows:

$$m = \frac{y_2 - y_1}{x_2 - x_1}. \tag{2}$$

If $c = 0$ in eq. 1 we can say that the two variables are directly proportional, however, if $c \neq 0$ the variables are <u>not</u> proportional.

Data that has a non-linear relationship is difficult to analyse. Therefore you must convert such data into straight-line form. This process is called linearisation. The "linearised" plot is then used to determine the dependence of the variables on each other.

Sometimes you will do experiments to investigate a relationship between two variables, which can only be determined by plotting a graph. In other experiments you will have to determine the value of a physical constant using a known relationship between two variables. In the latter case the constants are related to the slope or the intercept (or both) of a graph of data that you collect (and possibily linearise).

Tables 1 and 2 summarise the graphical shapes and corresponding mathematical equations that you already know. These are also the most likely graph shapes that your data will have if you plot them as is. Table 2 also has information on how to linearise data for each of the non-linear curves that your data may follow. (Note: These are not the only non-linear curves that data can have.)

Table 1: Straight line graphs, verbal and corresponding mathematical descriptions of relationship between dependent and independent variables.

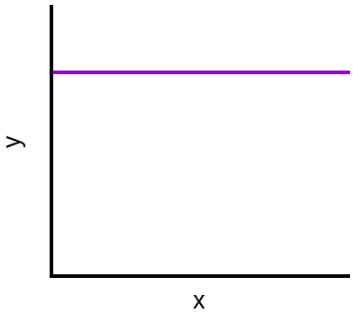
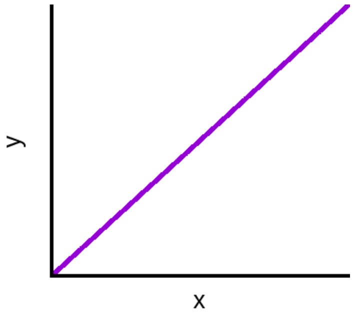
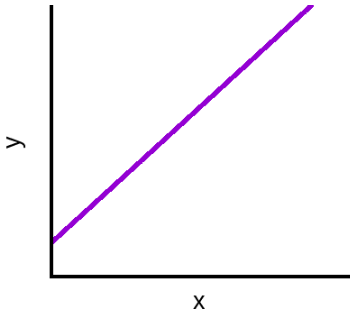
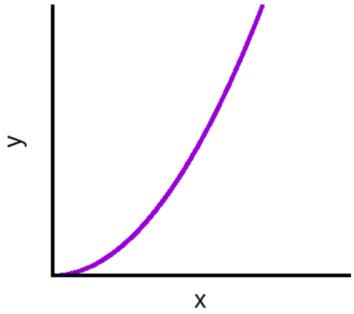
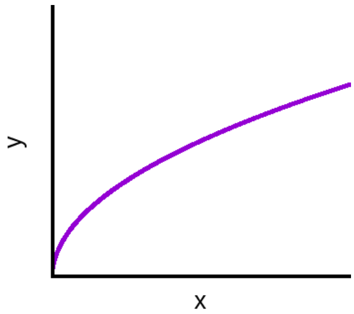
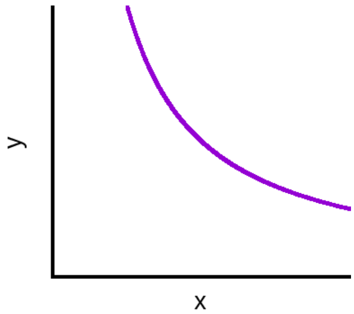
Graph shape	Relationship in words	Mathematical expression	Linearization
	No relationship between x and y. As x changes y remains unchanged.	$y = c$	none
	y increases as x increases. Both quantities approach zero simultaneously impliying that y is directly proportional to x	$y = mx$	Data is already linear. Draw best-fit line. Calculate slope (m)
	y increases as x increases. Non-zero intercept implies quantities not directly proportional	$y = mx + c$	Draw best-fit line. Calculate slope (m) and read intercept (c) off graph.

Table 2: Non-linear graph shapes, verbal and corresponding mathematical descriptions of relationship between dependent and independent variables, and how to linearize non-linear data.

Graph shape	Relationship in words and mathematically	Method for linearising data	Linearised equation (Your Goal!)
	y increases as the square of x $y = ax^2$	Calculate x^2 in a new column in your spreadsheet. Let $X = x^2$. Linearised eqn: $y = mX + c$ Plot y vs X.	$y = m\,(x^2) + c$
	y increases as the squareroot of x. $y = a\,x^{1/2}$	Calculate y^2 in a new column in your spreadsheet. Let $Y = y^2$. Plot Y vs x.	$y^2 = mx + c$
	y is inversely proportional to x. $y = ax^{-1}$	Calculate x^{-1} in a new column in your spreadsheet. Let $X = x^{-1}$. Plot y vs X.	$y = m\,(x^{-1}) + c$

3 Method for Linearising Data

1. Enter you data into a spreadsheet as you make measurements. Organise data in columns for the independent and dependent variable.

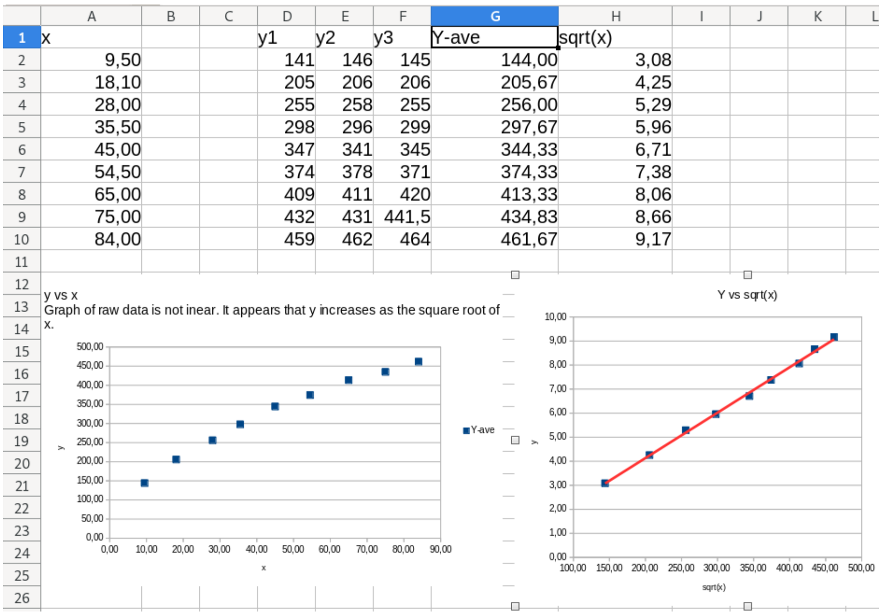
2. Have a separate column for each repeated measurements of the dependent variable for a chosen value of the independent variable. Calculate an average of the repeated quantity.

3. Plot a graph of the unlinearized data you collected. What shape is it? (See Tables 1 and 2 for guidlines for some common shapes.)

4. Linearize data appropriately.

5. Plot a graph of the linearized data. If you perform the linearisation correctly then the graph of the linearised quantities must be a straight line.

The method above is illustrated using Figure 2. The data shown in Figure 2 is from an experiment to find the relationship between 2 variables x (the independent variable) and y (the dependent variable). Plotting y versus x data points yields a non-linear curve in which y appears to increase as the square root of x. To linearise the data, the quantity $\sqrt{x}$ is calculated in the third column of the spreadsheet shown in Figure 2. A plot of y versus $\sqrt{x}$ yields a straight line graph. This confirms that y depends linearly on the square root of x.



x	y1	y2	y3	Y-ave	sqrt(x)
9,50	141	146	145	144,00	3,08
18,10	205	206	206	205,67	4,25
28,00	255	258	255	256,00	5,29
35,50	298	296	299	297,67	5,96
45,00	347	341	345	344,33	6,71
54,50	374	378	371	374,33	7,38
65,00	409	411	420	413,33	8,06
75,00	432	431	441,5	434,83	8,66
84,00	459	462	464	461,67	9,17

Figure 2: Data captured in a spreadsheet is at the top. Column H contains $\sqrt{x}$. The graph on the bottom left of the x and $y - ave$ data is non-linear. The graph on the right is of the linearised data, and the red line is a line of best fit through the data points.

If you think that the data ploted on the left in Figure 2 does not follow square root curve, see if one of the other linearisations given in Table 2 yields a straight line.